



Vallari Devang Ashar

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● ABOUT MYSELF

Computer Science and Engineering undergraduate specializing in Artificial Intelligence and Robotics, with research experience in multi-robot motion planning, embedded systems, and robotic simulation. Intern at ABB Robotics, with prior research and development experience at IIT Madras and DRDO.

● WORK EXPERIENCE

08/12/2025 - CURRENT - BENGALURU, INDIA

INTERN – ROBOTICS ABB ROBOTICS PVT. LTD., INDIA

- Evaluation and integration of CFree technology within RobotStudio for industrial robot applications.
- Exposure to offline robot programming, simulation workflows, and safety-critical robotics software.
- Collaboration with R&D engineers in ABB Robotics RA division.

20/08/2025 - 15/11/2025 - CHENNAI, INDIA

RESEARCH INTERN INDIAN INSTITUTE OF TECHNOLOGY (IIT), MADRAS

- Developed motion planning algorithms for multi-robot systems using Voronoi-based coverage optimization.
- Implemented 2D Voronoi partition coverage achieving 15–20% improvement in area coverage efficiency.
- Extended spanning-tree-based coordination methods for autonomous surveillance robots.
- Validated algorithms in simulation with 4–8 robots over large-scale environments.

15/05/2025 - 16/07/2025 - BENGALURU, INDIA

PROJECT INTERN DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION (DRDO)

- Verification and validation of mission-critical embedded systems for fighter aircraft programs at Combat Aircraft System Development and Integration Centre.
- Functional testing of flight control software on Xilinx Kintex-7 FPGA using VHDL.
- Experience with avionics protocols, real-time signal processing, and HIL testing.

15/03/2023 - 09/08/2025 - CHENNAI, INDIA

TEAM CAPTAIN TECHNOCRATS ROBOTICS

- Led robotics team in national and international competitions (DD Robocon, International Rover Challenge)
- Achieved 1st position nationally at DD Robocon Stage 1 (2025) and top-20 internationally at IRC 2025
- Produced high accuracy results for autonomous ball shooting and target centering using computer vision and image processing techniques on Jetson Nano.

19/10/2023 - 20/04/2024 - CHENNAI, INDIA

ROBOTICS RESEARCH INTERN ACK ROBOTICS

- Developed ROS2-based Ackermann steering robot using Nav2 and Gazebo.
- Implemented SLAM navigation with >90% path completion and <0.5 m localization error.
- Applied YOLOv8 and ResNet for industrial defect detection.

● PUBLICATIONS

Quantum-Enhanced Spiking Neural Networks for Closed-Loop Neuromodulation Systems 2024
IEEE Xplore Ashar, V., et al. Quantum-Enhanced Spiking Neural Networks for Closed-Loop Neuromodulation Systems. IEEE ICSES 2024.

Portable Welding Inspection System and Method For Detecting Structural Defects 2024
Intellectual Property Patent Filed — IN 202441097583.

● EDUCATION & TRAINING

19/09/2022 - Current - CHENAI , INDIA

B.TECH IN COMPUTER SCIENCE AND ENGINEERING(SPECIALIZATION AI & ROBOTICS) - VELLORE INSTITUTE OF TECHNOLOGY

Final grade: 8.76/10 | Level in EQF: 6 | Website: <https://chennai.vit.ac.in/>

01/04/2020 - 01/05/2022 - BENGALURU, INDIA

ALL INDIA SENIOR SCHOOL CERTIFICATE EXAMINATION (AISSCE) - 80%- CENTRAL BOARD OF SECONDARY EDUCATION

Field(s) of study: Basic programmes and qualifications

01/03/2020 - MUMBAI, INDIA

ALL INDIA SECONDARY SCHOOL EXAMINATION (AISSE) - 97.2%- CENTRAL BOARD OF SECONDARY EDUCATION

Field(s) of study: Basic programmes and qualifications

● SKILLS

Robotics

Experience with Robot Operating System (ROS) and Linux (ROS1 and ROS2) | Nav2 | MoveIt2 | Gazebo | RobotStudio-ABB | MATLAB and Simulink

Programming

Python | VHDL - Hardware Description Language | embedded systems | c, c++, c# | Machine Learning •

Tools

2D and 3D design software (Solidworks, AutoCad) | Git/Github, Docker, Gitlab | LaTeX

● PROJECTS

01/08/2025

Autonomous Crop Disease Detection Robot Autonomous ROS2-based mobile robot with Ackermann navigation, simulated in Gazebo, YOLO-based perception for real-time crop disease detection

01/06/2024

KIA - Open Source Robotic Arm 6-DOF manipulator with MoveIt2-based motion planning & GazeboSim.

01/02/2025

Mars Rover Abhimanyu(IRC) Designed off-board autonomy and control architecture using a space-grade antenna for long-range communication and planetary exploration tasks.

● VOLUNTEERING

01/01/2024 - 01/03/2024 Bishunpur, Jharkhand

Volunteer Teacher Taught Mathematics and English to underprivileged secondary school students.

● CERTIFICATIONS

21/06/2024 DeepLearning.AI (Stanford Online, Andrew Ng)

Machine Learning Specialization

05/08/2024 Microsoft

AI-900: Microsoft Azure AI Fundamentals

12/06/2024 Google

Foundations of Cybersecurity

Silverzone Foundation

iFLO - International French Language Olympiad Rank 1

- **LANGUAGE SKILLS**

Mother tongue(s): **GUJARATI**

Other language(s): **HINDI | ENGLISH | GERMAN | FRENCH | SPANISH**

- **HONOURS AND AWARDS**

SPACE ROBOTICS SOCIETY

CERTIFICATE OF ACHIEVEMENT INTERNATIONAL ROVER CHALLENGE 2025

- **STANDARDIZED TEST SCORES**

GRE

Q: 159 ,V:158, AWA:4.0

IELTS ACADEMIC

Band: 8.5